

ECON 131

PUBLIC ECONOMICS

Midterm 1 — Study Guide & Topic Outline

Spring 2026 • Prof. Danny Yagan • Exam: March 2, 5:10–6:10pm

EXAM FORMAT (based on 2024 midterm)

Section 1: True / False / Uncertain (6 statements × 2 pts = **12 pts**)

Section 2: Labor Supply Math Problem (~**16 pts**)

Section 3: Government Budget Math Problem (~**16 pts**)

Rules: No notes, no calculator, no phone. Written exam on paper. Show all work for partial credit.

PRIORITY KEY

HIGH PRIORITY Directly tested on 2024 midterm; core exam content	MEDIUM PRIORITY Likely appears in TF/U questions; important concepts
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LOW PRIORITY Background context; less likely to be directly tested

Course Map: What You Learned

The following 7 lectures constitute the full Midterm 1 scope. Lectures 4, 6, and 7 carry the heaviest exam weight.

#	Lecture	Gruber	Priority	What You Need to Know
L01	Intro to Public Economics	Ch. 1	LOW	Why govt intervenes; four questions of public finance; normative vs. positive; market failures; redistribution
L02	Theoretical Tools	Ch. 2	MEDIUM	Utility maximization, budget constraint, MRS, income/substitution effects, elasticity — the math toolkit for everything else
L03	Empirical Tools	Ch. 3	HIGH	Difference-in-differences, parallel pre-trends, RCTs, natural experiments — directly tested in TF/U

#	Lecture	Gruber	Priority	What You Need to Know
L04	Budget & Debt Analysis	Ch. 4	HIGH	Debt dynamics equation, primary deficit, Laffer curve, revenue-maximizing tax rate, sustainable debt condition
L05	Income Distribution & Poverty	Ch. 17–18	HIGH	Official vs. true poverty rate, CPI-RS, EITC, Gini coefficient, top income shares, inequality trends
L06	Tax Incidence	Ch. 19	HIGH	Economic vs. statutory incidence, elasticity rule, DWL formula, Danish high-earner study
L07	Optimal Labor Tax	Ch. 20–21	HIGH	Labor supply model (FOC, budget constraint), income/substitution effects, taxable income elasticity, optimal top rate concept

L01 Introduction to Public Economics	LOW PRIORITY
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Core question: When and how should the government intervene in the economy?

Four Questions of Public Finance

- When should the government intervene? → Market failures or redistribution needs
- How might the government intervene? → Tax/subsidize, mandate, provide directly, or finance
- What are the effects? → Direct effects (mechanical) vs. indirect effects (behavioral responses)
- Why does it intervene the way it does? → Political economy

Key Distinctions to Know

- **Normative vs. Positive:** Positive = how things are (empirical). Normative = how things should be (theoretical + values). Positive is a prerequisite for normative.
- **Market Failures:** Externalities, imperfect competition, asymmetric information, individual failures (behavioral).
- **Equity-Efficiency Trade-off:** Redistribution may reduce inequality but creates disincentives to work or save.

Exam relevance: This lecture is mostly background. You won't be asked to regurgitate it, but the vocabulary (normative/positive, market failures, direct/indirect effects) can appear in TF/U framing.

L02 Theoretical Tools	MEDIUM PRIORITY
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This lecture provides the math scaffold for everything else. You don't derive these from scratch on the exam — you use them inside the labor supply and tax problems.

Utility Maximization

- Consumer maximizes $U(X_1, X_2)$ subject to $p_1X_1 + p_2X_2 = Y$
- Solution condition: $MRS_{12} = p_1/p_2$ (marginal rate of substitution = price ratio)
- Equivalently: $MU_1/p_1 = MU_2/p_2$ (bang-per-buck equalized across goods)

$$\text{FOC: } dU/dX_1 - (p_1/p_2) \times dU/dX_2 = 0$$

Income & Substitution Effects (CRITICAL for L07)

- **Substitution effect:** Change in consumption due to relative price change, holding utility constant. A wage increase makes leisure relatively more expensive → work more.
- **Income effect:** Change in consumption due to change in purchasing power, holding relative prices constant. Higher wage makes you richer → 'buy' more leisure → work less.
- **Net effect on labor supply:** Ambiguous. If substitution > income, labor supply rises with wage. If income > substitution, labor supply falls.

Elasticity

- **Own-price elasticity:** $\varepsilon = (dQ/Q) / (dP/P) = (dQ/dP) \times (P/Q)$
- Inelastic: $|\varepsilon| < 1$. Elastic: $|\varepsilon| > 1$. Unit elastic: $|\varepsilon| = 1$
- Less elastic side of the market bears more of a tax — core incidence principle

L03 Empirical Tools

HIGH PRIORITY

This is directly tested in TF/U. Know the difference-in-differences setup cold.

Difference-in-Differences (DD)

The estimator: Compares the change in outcomes for a treatment group to the change in outcomes for a control group around a policy change.

$$DD = [Y(T, \text{After}) - Y(C, \text{After})] - [Y(T, \text{Before}) - Y(C, \text{Before})]$$

- **Key assumption — Parallel Pre-Trends:** In the absence of the policy, treatment and control groups would have trended the same way. You validate this by checking pre-period trends are parallel. It's an assumption, not proven by pre-trends alone.
- **What DD estimates:** The causal effect of the treatment, under the parallel trends assumption.

- **Threat to validity:** If something else changed at the same time that affected only the treatment group, DD is biased.

Other Empirical Methods

- **Randomized Controlled Trial (RCT):** Randomly assign treatment and control. Gold standard for causal inference. Practical limitations: compliance, SUTVA violations, external validity.
- **Natural Experiment:** A policy change or event that creates quasi-random variation in treatment. Examples: tax reforms affecting some income groups but not others.
- **Regression:** Controls for observables but cannot solve omitted variable bias on its own. DD is a regression-based method when implemented formally.

TF/U trap: “Parallel pre-trends validate DD” → TRUE. It is the required identifying assumption and the main empirical way to support DD’s validity.

L04 Budget & Debt Analysis

HIGH PRIORITY

The government budget problem appeared as a full ~16-point section in 2024. Know all formulas and the logic behind them.

Core Debt Dynamics Equation

MEMORIZE THIS: The evolution of government debt from year to year:

$$B(t+1) = (1 + r) \times B(t) + PD(t)$$

- **B(t):** Debt at start of year t
- **r:** Interest rate on debt
- **PD(t) = Primary Deficit = G(t) - T(t):** Government spending minus tax revenue, EXCLUDING interest payments
- **Total deficit:** $G(t) + r \times B(t) - T(t) = PD(t) + r \times B(t)$. Includes interest.

Tax Revenue and the Laffer Curve

If taxable income responds to the tax rate as $z(\tau) = (1 - \tau)^\phi$, then:

$$\text{Tax Revenue } TR(\tau) = \tau \times z(\tau) = \tau(1 - \tau)^\phi$$

The revenue-maximizing tax rate (top of the Laffer curve):

$$\tau^* = 1 / (1 + \phi)$$

- Higher ϕ (more elastic taxable income) $\rightarrow$ lower optimal revenue-maximizing rate
- Beyond τ^* , raising taxes actually loses revenue because behavioral responses dominate

Sustainable Debt

With constant B , G , τ across periods, solve $B(t+1) = B(t) = B$:

$$B = (1/r) \times [\tau(1-\tau)^{\phi} - G] = (TR - G) / r$$

- **Sustaining debt with a primary deficit:** Only possible if $r < g$ (interest rate below growth rate). If $r > g$ and $PD > 0$, debt grows without bound.
- **$r < g$ in practice:** Has held for much of U.S. history; allows some economists to argue deficits are more sustainable than the simple model implies.

Key U.S. Budget Facts

- Federal tax revenue $\approx$ 17–20% of GDP; Federal spending $\approx$ 21–24% of GDP $\rightarrow$ structural deficit
- U.S. debt-to-GDP $\approx$ 100% (2020s). Projected to keep rising.
- Main Federal taxes: Individual income tax (40%), payroll taxes (40%), corporate tax (15%)
- Main State taxes: Property (30%), sales/excise (30%), income/corporate (30%)

L05 Income Distribution & Poverty

HIGH PRIORITY

U.S. Income Inequality Trends

- Top 10% pre-tax income share: fell from $\sim$ 45% (1930s) to $\sim$ 33% (1970s), then rose to $\sim$ 50% (2018)
- Top 1%: about 20% of national income on pre-tax basis (up from $\sim$ 8% in 1980)
- Bottom 50% real income: grew only $\sim$ 1% from 1980–2014, while average grew 61%
- Capital income is more concentrated than labor income. Top 1% hold $\sim$ 40% of private wealth
- **Pre-tax vs. post-tax:** Taxes and transfers compress the distribution. Post-tax top shares are lower than pre-tax, but redistribution hasn't kept pace with pre-tax rise.

Poverty Measurement

- **Official poverty rate:** Compares pre-tax cash income to poverty threshold (set in early 1960s as $3 \times$ food budget). Updated annually by CPI.
- **Problem 1 — CPI overstates inflation:** Doesn't fully account for consumer substitution toward cheaper goods. CPI-RS corrects this $\rightarrow$ shows poverty fell more than official rate suggests.
- **Problem 2 — Excludes non-cash benefits:** Official rate ignores EITC, food stamps (SNAP), Medicaid. These programs reduced effective poverty significantly after 1972.

- **Bottom line:** Official poverty rate was roughly flat 1972–2005. Corrected measures (CPI-RS + EITC) show substantial decline. This is a classic TF/U “Uncertain” answer.

Key Transfers: EITC

- Earned Income Tax Credit: refundable tax credit that rises with earnings up to ~\$15,000, then phases out
- Unlike a lump-sum transfer, EITC depends on earnings → affects labor supply incentives
- Creates a phase-in range (subsidy to work), a flat range, and a phase-out range (implicit marginal tax)

Gini Coefficient

- Measures inequality on $[0, 1]$: 0 = perfect equality, 1 = one person has everything
- U.S. pre-tax Gini ≈ 0.5 . Post-tax/transfer ≈ 0.38 . European countries have lower post-tax Ginis.
- Limitation: two distributions can have the same Gini with very different shapes

L06 Tax Incidence	HIGH PRIORITY
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Tax incidence = who really bears the burden of a tax. Statutory incidence (who writes the check) $\neq$ economic incidence.

The Elasticity Rule

In a competitive market with supply elasticity $\epsilon_s > 0$ and demand elasticity $\epsilon_d < 0$:

$$\text{Consumer share} = \epsilon_s / (\epsilon_s - \epsilon_d)$$

$$\text{Producer share} = -\epsilon_d / (\epsilon_s - \epsilon_d)$$

- **Intuition:** The less elastic (more inelastic) side of the market bears more of the tax. They have fewer alternatives and can't avoid the tax as easily.
- **Perfectly inelastic demand:** Consumers bear 100% of the tax. Price rises by the full tax amount.
- **Perfectly elastic supply:** Consumers bear 100% (horizontal supply = perfectly competitive with constant costs).

Slope vs. Elasticity (Common TF/U Trap)

Slope $\neq$ elasticity! In a price-quantity graph, a steep (nearly vertical) demand curve = inelastic demand. A flat (nearly horizontal) demand curve = elastic demand.

2024 TF/U example: Supply slope = +1, demand slope = -1000. Demand is very steep (nearly vertical) → very inelastic → consumers bear most of the tax. Answer: TRUE.

Deadweight Loss (DWL)

A tax creates inefficiency by reducing quantity transacted below the efficient level:

$$DWL = -(1/2) \times [\epsilon_s \times \epsilon_d / (\epsilon_s - \epsilon_d)] \times (Q/P) \times (\Delta t)^2$$

- DWL rises with the square of the tax rate → large taxes are proportionally more costly than small taxes
- DWL rises with the absolute elasticities → more elastic goods have higher efficiency costs
- Implication: spread taxes across many goods to keep each rate low

Empirical Application: Denmark High-Earner Study (Kleven et al.)

- 1991 Danish tax reform gave a tax break to high-earning foreigners choosing to reside in Denmark
- Result: number of qualifying high-earning foreigners rose sharply relative to non-qualifying earners
- **Conclusion:** High earners DO respond to taxes in choosing where to live. Significant mobility response.
- **TF/U trap:** “Denmark shows high earners don’t respond to taxes” → FALSE

L07 Optimal Labor Tax & Labor Supply

HIGH PRIORITY

This is almost certainly the math section of the exam. The 2024 midterm used an identical model structure. Understand every step.

The Standard Model Setup

- **Utility:** $U(c, l) = c - l^{1+\eta} / (1 + \eta)$
- **Budget constraint:** $c = (1 - t)wl + T$ (post-tax wage times labor, plus lump-sum transfer)
- **Tax system:** Flat rate t on earnings; lump-sum transfer T to all households

Solving for Optimal Labor Supply

Substitute budget constraint into utility and take FOC with respect to l :

$$\begin{aligned} \text{FOC: } (1-t)w - l^{1/\eta} &= 0 \\ \Rightarrow l^* &= [(1-t)w]^\eta \end{aligned}$$

Optimal consumption:

$$c^* = (1-t)w \times l^* + T = [(1-t)w]^{(1+\eta)} + T$$

Taxable Income Elasticity

Elasticity of l^* with respect to the post-tax wage $(1-t)w$:

$$\begin{aligned}\varepsilon &= (dl^*/d[(1-t)w]) \times [(1-t)w / l^*] \\ &= \eta \times [(1-t)w]^{(\eta-1)} \times [(1-t)w / (1-t)w]^\eta \\ &= \eta\end{aligned}$$

Key result: In this model, the taxable income elasticity equals η , the curvature parameter. Empirically, for top earners this is estimated at 0.1–0.5 (well below 1).

Income Effect of Lump-Sum Transfer

- $l^* = [(1-t)w]^\eta$ does NOT depend on T
- Therefore: elasticity of labor supply w.r.t. $T = 0$
- **Implication:** Income effects DO NOT dominate substitution effects in this model. The substitution effect (post-tax wage) determines labor supply; the income effect from the transfer does not.

Labor Market Equilibrium

If labor demand is $l^d(w) = w^{(-\eta^d)}$, set $l^* = l^d$ and solve for w :

$$\begin{aligned}[(1-t)w]^\eta &= w^{(-\eta^d)} \\ w^* &= (1-t)^{(-\eta / (\eta + \eta^d))}\end{aligned}$$

- Tax increase (higher t) $\rightarrow$ lower post-tax wage $\rightarrow$ lower labor supply $\rightarrow$ lower equilibrium wage
- How much wage falls depends on both supply elasticity (η) and demand elasticity (η^d)

Optimal Top Tax Rate (Concept Only — Don't Memorize Formula)

- **Key insight:** The optimal top tax rate is lower when the taxable income elasticity is higher, because high taxes cause more behavioral distortion.
- If η is large, top earners strongly reduce earnings in response to tax increases $\rightarrow$ revenue actually falls
- Utilitarian social welfare: with no behavioral response, optimal is to fully equalize consumption (100% marginal rate). With response, must trade off equity gains vs. efficiency costs.
- **What Yagan tests:** The logic ("optimal top rate is small if elasticity is large"), not the formula itself.

EITC vs. Lump-Sum Transfer

- EITC: transfer depends on earnings level $\rightarrow$ affects marginal incentive to work in each range

- Lump-sum transfer T: same amount regardless of earnings → only income effect, no substitution effect on the margin
- U.S. transfers (EITC, SNAP, TANF) are all earnings-contingent, unlike the model's lump-sum T

True / False / Uncertain — Master Table

These are modeled on exact 2024 exam questions plus high-probability analogues from the lectures. For every TF/U answer, you must give the verdict PLUS the key mechanism.

Statement	Verdict	Key Reasoning
A tax rate cut should always make people work more.	FALSE	<i>Income effect can offset substitution effect. If income > substitution, labor supply falls.</i>
Parallel pre-trends validate the difference-in-differences estimator.	TRUE	<i>Parallel pre-trends is exactly the identifying assumption DD requires.</i>
The official U.S. poverty rate was flat from 1972–2005.	UNCERTAIN	<i>Official rate: roughly flat. CPI-RS + EITC corrections show substantial decline.</i>
If demand slope = -1000 and supply slope = $+1$, consumers bear most of the tax.	TRUE	<i>Steep demand = inelastic. Less elastic side bears more incidence.</i>
If earnings don't respond to taxes, a utilitarian government equalizes consumption.	TRUE	<i>No efficiency cost of redistribution → optimal to fully equalize (diminishing MU).</i>
Denmark shows that high earners don't decide where to live based on taxes.	FALSE	<i>Kleven et al.: 1991 Danish tax break caused sharp increase in high-earner foreigners.</i>
The primary deficit equals total government spending minus tax revenue.	FALSE	<i>Primary deficit = $G - T$, EXCLUDING interest payments. Total deficit = $G + rB - T$.</i>
A tax on a perfectly inelastic good creates zero deadweight loss.	TRUE	<i>No behavioral response → no quantity reduction → no DWL triangle.</i>
The revenue-maximizing tax rate is always 50%.	FALSE	<i>$\tau^* = 1/(1+\phi)$. Only equals 50% when $\phi = 1$. Depends on behavioral response elasticity.</i>
Sustaining positive debt with a primary deficit requires $r < g$.	TRUE	<i>If $r > g$ and $PD > 0$, debt grows without bound. $r < g$ is the necessary condition.</i>

Statement	Verdict	Key Reasoning
The EITC is equivalent to a lump-sum transfer to low-income workers.	FALSE	EITC depends on earnings, creating phase-in/phase-out incentive effects. Lump-sum doesn't.
Top U.S. pre-tax income shares are higher today than in 1980.	TRUE	Top 1% share rose from ~8% to ~20% of national income from 1980 to today.

Formula Reference Sheet

You need to know these. Per syllabus: you don't need to memorize formulas derived in class, but you do need the concepts and lessons they communicate. However, for budget and labor supply problems, you must be able to derive them from scratch.

Concept	Formula	Note
Budget constraint (household)	$c = (1-t)wl + T$	Tax rate t , wage w , labor l , transfer T
Optimal labor supply (given model)	$l^* = [(1-t)w]^\eta$	FOC of the standard utility model
Optimal consumption	$c^* = [(1-t)w]^{(1+\eta)} + T$	Plug l^* into budget constraint
Taxable income elasticity	$\varepsilon = \eta$	In the standard model; measures labor supply response to post-tax wage
Gov't debt evolution	$B(t+1) = (1+r)B(t) + PD(t)$	PD = primary deficit = $G - T$
Tax revenue (Laffer setup)	$TR(\tau) = \tau(1-\tau)^\phi$	Taxable income = $(1-\tau)^\phi$
Revenue-maximizing tax rate	$\tau^* = 1/(1+\phi)$	Peak of Laffer curve
Sustainable debt level	$B = (TR - G) / r$	Constant B, G, r ; requires $r > 0$ and $TR > G$
Consumer tax incidence share	$\varepsilon_s / (\varepsilon_s - \varepsilon_g)$	$\varepsilon_s > 0, \varepsilon_g < 0$
Deadweight loss	$-(1/2) [\varepsilon_s \varepsilon_g / (\varepsilon_s - \varepsilon_g)] (Q/P) (\Delta t)^2$	Rises with square of tax rate and with elasticities
Equilibrium wage (labor mkt)	$w^* = (1-t)^{-\eta / (\eta + \eta^d)}$	Set labor supply = labor demand and solve

Study Plan for Today

You have limited hours before the exam. Here's how to allocate your time for maximum return.

#	Time	Topic	Priority	What to do
1	30 min	Labor supply model	HIGH	Re-derive l^* from scratch. Compute the taxable income elasticity by hand. Then solve for equilibrium wage. Do it without notes.
2	25 min	Government budget problem	HIGH	Re-derive debt dynamics equation. Compute τ^* , maximum revenue, and sustainable B . Do the algebra without notes.
3	20 min	Past midterm (2024)	HIGH	Work through all three sections with the exam PDF. Check your answers against the solutions. Identify where you lose partial credit.
4	15 min	TF/U master table	HIGH	Cover the Verdict and Reasoning columns. State each answer out loud with a 2-sentence justification. Flag anything you can't justify.
5	15 min	Tax incidence & DWL	MEDIUM	Practice the elasticity rule. Given supply/demand slopes, identify who bears more of the tax. Know the DWL logic (rises with t^2 and elasticity).
6	15 min	Poverty & inequality facts	MEDIUM	Know: official vs. corrected poverty rate, CPI-RS, EITC structure, Gini range, top 1% share (20%). These are TF/U targets.
7	10 min	Empirical tools	MEDIUM	Recite the DD formula. State the parallel pre-trends assumption. Name one threat to validity.
8	10 min	Buffer / review weak areas	varies	Use any remaining time on whichever topic felt shakiest. Intro (L01) is low-priority; skip if time is short.

EXAM STRATEGY

TF/U (Section 1): Two sentences per answer. Verdict first, mechanism second. "Uncertain" is valid — but you still need to defend a direction.

Math Sections: Show all steps. Set up the Lagrangian or directly substitute the budget constraint. Take FOC. Solve explicitly. Partial credit is generous.

Time: ~60 minutes for 3 sections. Spend ~15 min on TF/U, ~20 min on each math section. Don't get stuck — move on and return.

Units: Label your variables. State what each symbol represents. Graders reward clarity.